

1. (3 pts) Choose the best option to describe each matrix below. Your choices are:

- Reduced echelon form
- Echelon form but not reduced echelon form
- Not echelon form.

a) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

not echelon

b) $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Echelon

c) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Not echelon

2. (2 pts) Either explain carefully in words or show what you would do for the first two steps in row reducing the matrix A without using a calculator.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Replace R_2
by $R_2 - 2R_1$

Replace R_3
by $R_3 - R_2$

other steps
are possible

3. (3 pts) Is the vector $\begin{bmatrix} 7 \\ -9 \\ 12 \end{bmatrix}$ in the span of the vectors $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$

Yes

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & -3 & -9 \\ 1 & 3 & 12 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ -9 \\ 12 \end{bmatrix} = -3 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$$

On the other version
of the quiz the
vector $\begin{bmatrix} 7 \\ -9 \\ 10 \end{bmatrix}$ is
not in the span

(4 pts) Given that $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$ write the matrix equation $A\mathbf{x} = \mathbf{b}$

a) as a system of equations

$$\begin{aligned} x_1 + 2x_2 &= 10 \\ x_1 + 3x_2 &= 20 \\ x_1 + 4x_2 + x_3 &= 30 \end{aligned}$$

b) as a vector equation.

$$x_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$$

5. (3 pts) Describe the **type** of solution set for the system $A\mathbf{x} = \mathbf{b}$ given the **augmented** matrix

$[A \ \mathbf{b}]$

a) $\begin{bmatrix} 2 & 3 & 1 & | & 2 \\ 0 & 1 & 4 & | & 6 \\ 0 & 0 & 0 & | & 5 \end{bmatrix}$

No solution
Since
pivot in
augmented
column

b) $\begin{bmatrix} 2 & 3 & 1 & | & 2 \\ 0 & 1 & 4 & | & 6 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$

infinite
solution set
since x_3
is a free
variable

c) $\begin{bmatrix} 1 & 2 & 1 & | & 4 \\ 0 & 2 & 1 & | & 6 \\ 0 & 0 & 1 & | & 0 \end{bmatrix}$

unique solution
since the system
is consistent
with a pivot in
every column of A .
No free variables